

ANNU MATHUR

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EDUCATION

National Institute of Technology, Srinagar

B.Tech in Computer Science and Engineering | CGPA: 7.64

Aug 2023 – Aug 2027

Srinagar, India

WORK EXPERIENCE

Winter Research Intern, DIAT Pune (Govt. of India, MoD)

Dec 2025 – Jan 2026

Dr. B.R. Ambedkar 3D AI Avatar (RAG System) | 15-Member Team | Guided by Prof. C.R.S Kumar

[Live Demo](#) | [Certificate](#)

- Contributed to the backend of a 3D AI avatar in a 15-member team using FastAPI, Python, Google Gemini, and REST APIs.
- Built the RAG pipeline by cleaning documents, splitting them into chunks, and retrieving relevant content using BM25 keyword search and Qdrant vector search.
- Added API key fallback and rate-limit handling to keep the AI service working during live use.
- Deployed the backend on Render and frontend on Vercel, and used MongoDB Atlas to store user questions and AI responses.

PROJECTS

Customer Shopping Behavior Analysis | Python, MySQL, Pandas, Matplotlib, Seaborn

[GitHub](#)

- Analyzed 3,900+ customer records using Python and Pandas to study spending, discounts, customer segments, and product trends.
- Wrote 15 MySQL queries using joins, CTEs, and window functions to answer questions on revenue, discounts, loyalty, and top products.
- Created charts using Matplotlib and Seaborn to compare spending, sales by category, age groups, seasons, and customer behavior.

WanderNest – Vacation Rental Platform | Node.js, Express.js, MongoDB Atlas, Cloudinary, Passport.js

[GitHub](#)

- Built a full-stack vacation rental platform with user authentication, property listings, reviews, image uploads, and location-based features.
- Implemented user authentication and authorization using Passport.js and sessions, with MongoDB Atlas for storing users, listings, reviews, and sessions.
- Added Cloudinary for image uploads, Mapbox for location coordinates, and Joi for server-side validation of listing and review data.

OPEN SOURCE CONTRIBUTION

Eventra – GirlScript Summer of Code 2026 | React.js, JavaScript, Git, GitHub

[Pull Request](#)

- Fixed responsive layout issues and refined form validation on the Feedback and Contact pages, removing duplicated UI components in the process.
- Resolved merge conflicts and addressed maintainer review feedback to get the change merged (PR #991).

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL

Backend: FastAPI, Node.js, Express.js, REST APIs

Frontend: React.js, HTML5, CSS3, Bootstrap

Databases: MySQL, MongoDB Atlas

Cloud & DevOps: Render, Vercel, GitHub Actions

Tools: Git, GitHub, Postman, VS Code, IntelliJ IDEA, Cloudinary

Core Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

LEADERSHIP EXPERIENCE

Core Team Member — Range-Chinar Fest

Oct 2025 — [Certificate](#)

- Managed registration and payment verification for **1,000+ participants**, ensuring smooth check-in and accurate record management.
- Coordinated with volunteers and organizing teams to manage participant check-in and support event operations.

ACHIEVEMENTS & CERTIFICATIONS

LeetCode: 270+ problems solved, Contest Rating **1711** (Top 13.06%)

[Profile](#)

DSA with Java — Apna College

[Certificate](#)

Full-Stack Web Development — Apna College

[Certificate](#)

Java (Basics) — HackerRank

[Certificate](#)